Markscheme

Mathematics: analysis and approaches

Practice question bank

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17. (a)

Consider the function $f(x) = \dfrac{2x+10}{(x-3)(x+1)}$.

Write $f(x)$ as the sum of partial fractions in the form $\dfrac{A}{x-3} + \dfrac{B}{x+1}$.

$$2x + 10 = A(x+1) + B(x-3)$$ *(M1)*

substituting $x = 3$: $16 = 4A \Rightarrow A = 4$ **A1**

substituting $x = -1$: $8 = -4B \Rightarrow B = -2$ **A1**

[3 marks]